EXPERIMENT 27

NAME: B.Akshay Kumar Reddy

Reg no: 192412246

CODE:

```
clc;
clear;
close all;
% Processing Gain
Rb = 1000;        % Data Rate (bps)
Rc = 10000;       % Chip Rate (chips/s)
PG = Rc/Rb;
PG_dB = 10*log10(PG);
fprintf('Processing Gain = %.2f\n', PG);
fprintf('Processing Gain (dB) = %.2f dB\n', PG_dB);
% Generate PN Sequence
 % 31-bit PN sequence
pn = randi([0 1],1,31);
% Display
disp('PN Sequence:');
disp(pn);
% Plot
figure;
stem(pn,'filled');
title('Pseudo Noise (PN) Sequence');
xlabel('Chip Number');
ylabel('Chip Value');
```

grid on;

OUTPUT;

